

Report Assignment 4b

Group 2

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1 Introduction

Risk-based allocation strategies seek to construct portfolios using only the covariance structure of asset returns, abandoning expected returns altogether. The motivation is well known: in the classical Markowitz framework, expected returns are the dominant source of estimation error and produce extreme, unstable weights. Discarding them yields portfolios that are more robust out of sample, more interpretable, and easier to defend in a real-money context.

In this report we study two representative members of this family on the Euro Stoxx 50 universe: the *Equal Risk Contribution* (ERC) portfolio and the *Hierarchical Risk Parity* (HRP) procedure of López de Prado. Both are evaluated with monthly rebalancing and a two-year rolling estimation window, and all analyses are run side-by-side under three covariance estimators: the sample covariance, the constant-correlation shrinkage of Ledoit and Wolf (2003) and the single-factor (market) shrinkage of Ledoit and Wolf (2002), with the equally-weighted return of the surviving universe as the market proxy.

2 Part I: Equal Risk Contribution (ERC)

The Equal Risk Contribution (ERC) portfolio seeks weights $\mathbf{w} \in \mathbb{R}_+^N$, with $\mathbf{1}^\top \mathbf{w} = 1$, such that every asset contributes equally to the total portfolio volatility.

Let $\sigma(\mathbf{w}) = \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$ be the portfolio volatility. Because σ is homogeneous of degree 1, Euler's decomposition gives

$$\sigma(\mathbf{w}) = \sum_{i=1}^N w_i \frac{\partial \sigma(\mathbf{w})}{\partial w_i} = \sum_{i=1}^N RC_i, \quad RC_i = w_i \frac{(\Sigma \mathbf{w})_i}{\sigma(\mathbf{w})}. \quad (1)$$

The ERC condition requires $RC_i = RC_j$ for all pairs (i, j) , which together with (1) implies $RC_i = \sigma(\mathbf{w})/N$ for every asset.

2.1 Implementation

Because no closed-form solution exists in the presence of correlations, we cast the ERC condition as a nonlinear least-squares programme. Defining the *un-normalised* risk contribution of asset i as

$$\widetilde{RC}_i(\mathbf{w}) = w_i (\Sigma \mathbf{w})_i, \quad (2)$$

the ERC objective minimises the pairwise variance of the $\{\widetilde{RC}_i\}$:

$$\mathcal{L}(\mathbf{w}) = \sum_{i=1}^N \sum_{j=1}^N \left[\widetilde{RC}_i(\mathbf{w}) - \widetilde{RC}_j(\mathbf{w}) \right]^2. \quad (3)$$

An equivalent, numerically convenient reformulation is

$$\mathcal{L}(\mathbf{w}) = N \|\tilde{\mathbf{r}}\mathbf{c}\|^2 - (\mathbf{1}^\top \tilde{\mathbf{r}}\mathbf{c})^2, \quad (4)$$

where $\tilde{\mathbf{r}}\mathbf{c} = (\widetilde{RC}_1, \dots, \widetilde{RC}_N)^\top$. This identity follows from $\sum_i \sum_j (a_i - a_j)^2 = 2N \sum_i a_i^2 - 2(\sum_i a_i)^2$ and an overall factor of $1/2$ absorbed in the constant. Our implementation directly uses (4), coded as `erc_objective_function`. The objective is minimised subject to the constraints:

$$\mathbf{1}^\top \mathbf{w} = 1, \quad w_i \geq 0 \quad \forall i, \quad (5)$$

using `scipy.optimize.minimize` with the SLSQP method. The solver is initialized at the *inverse-volatility* (IV) portfolio,

$$w_i^{\text{IV}} = \frac{\sigma_i^{-1}}{\sum_{j=1}^N \sigma_j^{-1}}, \quad \sigma_i = \sqrt{\Sigma_{ii}}, \quad (6)$$

which is the ERC solution when Σ is diagonal.

After optimisation the weights are renormalized, and a post-hoc check verifies that the percentage risk contributions (PCR) satisfy

$$\left| \frac{RC_i}{\sigma(\mathbf{w})} - \frac{1}{N} \right| < \delta \quad \forall i, \quad \delta = 10 \text{ bps}. \quad (7)$$

A warning is printed if the tolerance is breached.

2.1.1 Risk-contribution calculation

For diagnostic purposes, absolute risk contributions are computed in `risk_contribution`. Given a matrix $W \in \mathbb{R}^{P \times N}$ of P portfolio weight vectors, the function returns

$$RC_{p,i} = \frac{w_{p,i} (\Sigma \mathbf{w}_p)_i}{\sqrt{\mathbf{w}_p^\top \Sigma \mathbf{w}_p}}, \quad p = 1, \dots, P, \quad i = 1, \dots, N, \quad (8)$$

2.2 Sanity Checks: Risk-Contribution Dispersion and the IV–ERC Equivalence

2.2.1 IV–ERC Equivalence

If Σ is diagonal, the ERC solution is the inverse-volatility portfolio: when Σ is diagonal we have $(\Sigma \mathbf{w})_i = \sigma_i^2 w_i$, so

$$RC_i = w_i \frac{\sigma_i^2 w_i}{\sigma(\mathbf{w})} = \frac{w_i^2 \sigma_i^2}{\sigma(\mathbf{w})}.$$

The ERC condition $RC_i = RC_j$ yields $w_i^2 \sigma_i^2 = w_j^2 \sigma_j^2$, i.e. $w_i \sigma_i = w_j \sigma_j$ (positive weights), so $w_i \propto \sigma_i^{-1}$. Normalizing gives $w_i^* = \sigma_i^{-1} / \sum_j \sigma_j^{-1}$.

To verify this equivalence empirically, for each rebalance date we:

1. Extract the sample covariance matrix $\hat{\Sigma}$ for the current universe.
2. Set all off-diagonal entries to zero, producing the diagonal matrix $\hat{\Sigma}_{\text{diag}}$.
3. Compute the IV portfolio on $\hat{\Sigma}$ and the ERC portfolio on $\hat{\Sigma}_{\text{diag}}$.
4. Record the ℓ_1 distance between the two weight vectors: $\delta^{\text{IV-ERC}} = \|\mathbf{w}^{\text{IV}} - \mathbf{w}^{\text{ERC}}\|_1$.

	$\ \mathbf{w}^{\text{IV}} - \mathbf{w}^{\text{ERC}}\ _1$
Mean	1.05×10^{-16}
Max	4.48×10^{-16}

Table 1: ℓ_1 distance between IV and ERC weights.

All values are well below machine epsilon ($\approx 2.22 \times 10^{-16}$), confirming that the two strategies are numerically identical when Σ is diagonal, in agreement with the analytical result derived above.

2.2.2 Risk-Contribution Dispersion

The second sanity check verifies that the ERC optimizer successfully equalizes risk contributions when applied to the *full* (non-diagonal) sample covariance matrix $\hat{\Sigma}$. For each rebalance date we compute the ERC weights $\mathbf{w}^{\text{ERC}}$, retrieve the absolute risk contributions RC_i , and measure their dispersion via the normalized range:

$$\Delta_t = \frac{\max_i RC_{i,t} - \min_i RC_{i,t}}{\sum_i RC_{i,t}}. \quad (9)$$

We also track the mean percentage risk contribution (PCR) at each rebalance and compare it to the theoretical target $1/N_t$, where N_t is the number of assets in the universe at time t .

Normalized range Δ_t	
Mean	1.59×10^{-2}
Max	2.00×10^{-2}

Table 2: Dispersion of risk contributions across rebalances (full $\hat{\Sigma}$).

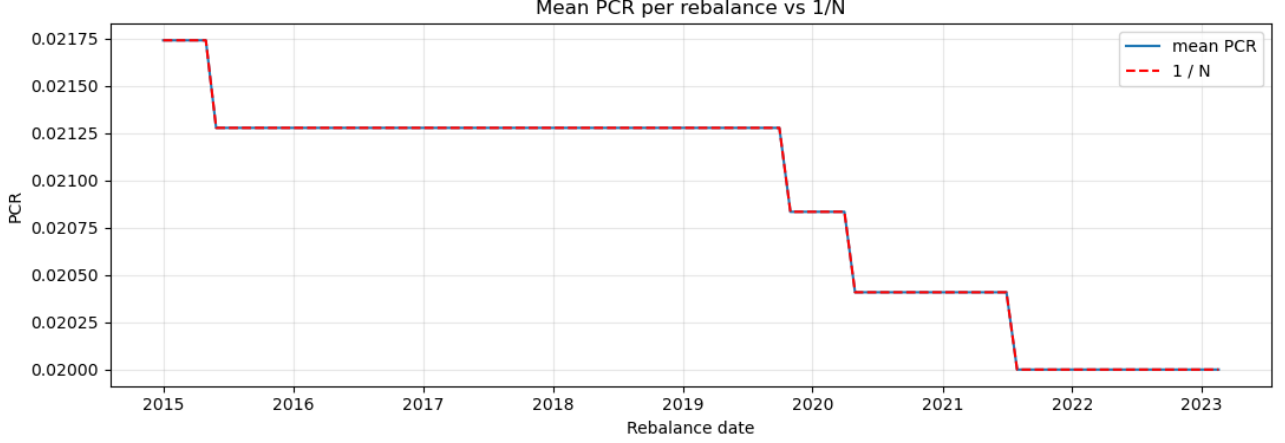


Figure 1

Figure 1 plots the mean PCR at each rebalance alongside the theoretical target $1/N_t$. The two series overlap perfectly throughout the entire sample, with the step-downs reflecting reductions in universe size N_t over time.

2.3 Backtest Across Covariance Estimators

2.3.1 Setup

We backtest the ERC strategy on the Euro Stoxx 50 universe with monthly rebalancing on the last trading day of each month, a trailing estimation window of 2 years of trading days, and a minimum coverage filter of 95%. At each rebalance date, ERC weights are computed by calling `equal_risk_contribution_portfolio` on each of the three covariance estimates: sample ($\hat{\Sigma}$), constant-correlation shrinkage ($\hat{\Sigma}_{cc}$), and market-factor shrinkage ($\hat{\Sigma}_{mf}$). The weight schedule is then passed to `backtest` and `portfolio_returns` to produce the daily P&L series. We report four metrics: cumulative return, 63-day EWM annualised realised volatility, drawdown, and monthly turnover

2.3.2 Results

Estimator	Total return	Ann. return	Avg. ann. vol	Max drawdown	Avg. turnover
Sample	100.28%	8.75%	17.59%	−39.5%	1.41%
Constant corr.	100.28%	8.75%	17.59%	−39.5%	1.41%
Market factor	100.28%	8.75%	17.59%	−39.5%	1.41%

Table 3: ERC backtest summary across covariance estimators.



Figure 2: ERC backtest across the three covariance estimators: cumulative return (top-left), 63-day EWM annualised realised volatility (top-right), drawdown (bottom-left), and monthly turnover (bottom-right).

2.3.3 Discussion

Why estimator choice is irrelevant for ERC. The most striking feature of Table 3 and Figure 2 is the *complete coincidence* of all three estimators across every metric: the three lines are visually indistinguishable in all four panels. This is not a numerical accident, it reflects a fundamental property of the ERC strategy. Unlike mean-variance optimisation, which inverts Σ and is therefore acutely sensitive to its eigenspectrum, ERC only uses Σ to *evaluate* risk contributions and balance them. The optimisation objective $\mathcal{L}(\mathbf{w}) = \sum_{i,j} (\bar{RC}_i - \bar{RC}_j)^2$ has a unique interior minimum that depends on the *correlation structure* of Σ , not on its precise scale.

Turnover. Average monthly one-way turnover is 1.41% across all three estimators, which is low by active-management standards. The turnover is not constant: spikes occur at universe-change events (when a stock enters or exits the eligible universe) and during high-volatility regimes such as March 2020, where the spike reaches $\approx 10\%$, as the abrupt shift in the covariance structure forces a significant reallocation of risk budgets.

Transaction costs. Given the low average turnover and the monthly frequency of portfolio rebalances, transaction costs have a negligible impact, as we can see from the following table:

	Value
Total return (gross)	100.28%
Total return (net)	100.04%
Total cost drag	0.24%
Annualised cost drag	0.03%

Table 4: ERC sample estimator — gross vs. net of 5 bps transaction costs.

3 Part II: Hierarchical Risk Parity (HRP)

The Hierarchical Risk Parity (HRP) procedure is an allocation strategy designed to address "Markowitz's Curse," referring to the extreme instability and noise sensitivity of quadratic optimizers when assets are highly correlated. Unlike traditional mean-variance frameworks, HRP does not require the inversion of the covariance matrix, ensuring robust solutions even when the matrix is singular or ill-conditioned.

The implementation follows a structured three-step mathematical pipeline:

1. **Distance Metric Construction:** The process begins by converting the covariance matrix into a correlation matrix ρ . Since raw correlation is not a formal metric, it is transformed into a distance measure using the formula:

$$d_{i,j} = \sqrt{0.5(1 - \rho_{i,j})}$$

This transformation is essential as it satisfies the three metric axioms—non-negativity, symmetry, and the triangle inequality—providing a consistent foundation for the clustering algorithm.

2. **Hierarchical Clustering:** Agglomerative clustering is performed using the `scipy.cluster.hierarchy.linkage` function. The procedure is parameterized by:
 - **Linkage Method:** Defines the methodology to calculate distances between clusters and other nodes. In this study, we utilize at most the **Ward** linkage, which aims to minimize the increase in total within-cluster variance at each merge step. For instance, when considering two potential clusters for a merge, Ward’s method selects the pair that results in the minimum increase in the sum of squared distances to the new cluster’s centroid, effectively creating more compact and balanced groups compared to other methods.
 - **Distance Metric:** While the linkage is built upon the correlation-derived distance, the internal distance calculation between rows for the linkage matrix is fixed to **Euclidean**. It is important to note that if a custom distance metric were required, the documentation specifies that the input must be provided as a pre-computed distance matrix, transformed into a condensed one-dimensional vector representing the upper triangle of that matrix.
3. **Quasi-Diagonalization and Iterative Weighting:** The final stage organizes the assets and allocates capital:
 - **Quasi-Diagonalization:** The rows and columns of the covariance matrix are reordered based on the "leaf" order of the dendrogram. This reorganization places highly correlated assets in proximity along the diagonal, revealing the block-diagonal structure and isolating noisy off-diagonal elements, as illustrated in Figure 3.
 - **Recursive Allocation:** Weights are calculated iteratively through a top-down traversal of the tree using either **Recursive Bisection** (halving the leaves regardless of height) or **Dendrogram Iteration** (splitting based on merge distances). Both methods apply Inverse Variance Parity (IVP) within clusters to allocate more capital to lower-variance nodes, bypassing full matrix inversion.

3.1 Correlation Detoning

The detoning procedure serves as a preprocessing step for hierarchical clustering and consists of removing the first principal component from the correlation matrix. As observed in previous assignments, the first eigenvalue of an empirical correlation matrix typically represents the "market factor".

By subtracting this rank-1 market component and rescaling the diagonal back to unity, the procedure effectively "silences" the most pervasive source of co-movement. Consequently, the resulting clusters are expected to reflect more accurate structural similarities between assets rather than being contaminated by general market trends.

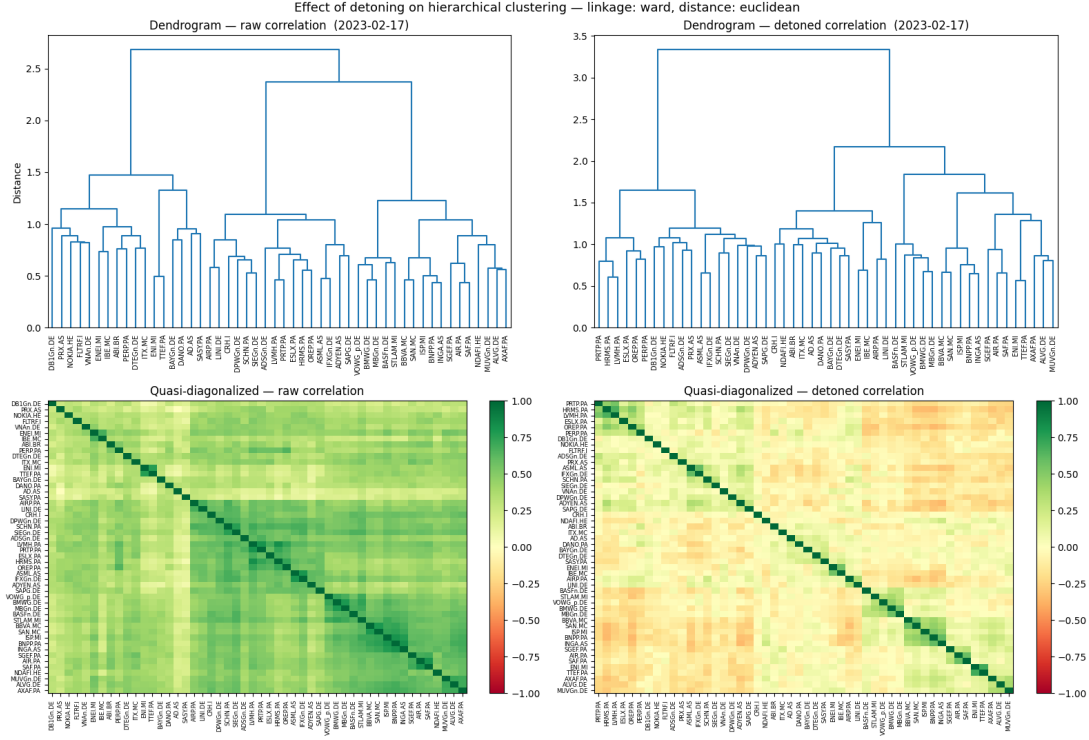


Figure 3: Dendrogram and Correlation visual inspection after correlation detoning (Last rebalance date).

The visualizations in Figure 3, obtained from the final rebalance date, demonstrate the efficacy of the HRP pipeline. The quasi-diagonalized correlation matrix shows that the algorithm has worked perfectly in grouping highly correlated assets along the diagonal. By isolating these groups, the noisy off-diagonal blocks become visually and mathematically separated, facilitating a more robust allocation.

The dendrogram in Figure 3 further illustrates this hierarchical organization. The branch heights encode the merge distances, providing a clear map of asset relationships once the pervasive market noise has been removed. This tree structure is essential for the subsequent allocation phase, as it defines the precise sequence of clusters used for the top-down capital distribution.

3.2 Volatility Allocation Logic

The final stage of the HRP pipeline involves a top-down traversal of the hierarchical structure to distribute capital based on the risk characteristics of the identified clusters. The fundamental objective is to allocate more capital to the lower-variance branch of each split, thereby achieving a more robust risk balancing compared to traditional methods.

3.2.1 Comparison of Traversal Approaches

The way the tree is traversed dictates which assets are compared against each other, significantly impacting the final weight distribution:

- **Recursive Bisection:** Implemented via the `recursive_bisection` function, this approach follows the logic originally proposed by Lopez de Prado. It ignores the actual merge distances (heights) in the dendrogram and simply bisects the ordered list of leaves. This results in a balanced binary tree where the portfolio is halved at each step until individual assets are reached.
- **Dendrogram Iteration:** This method, performed by the `dendrogram_iteration` function, strictly follows the hierarchy represented in the linkage matrix. At every step, the algorithm identifies and splits the cluster with the largest merge distance. This approach respects the actual statistical groupings found during clustering, though it may result in an asymmetric tree if the underlying data structure is not naturally balanced.

3.2.2 Logic and Nested List Processing

To manage the hierarchical allocation, the code utilizes a **nested list** structure to represent the branches of the tree. This structure is processed by the `top_down_allocation` function through the following logic:

1. **Flattening:** The function uses `flatten_list` to extract all asset labels within a specific branch, allowing the retrieval of the relevant sub-covariance matrix.
2. **Cluster Variance:** The variance of each branch is computed via `get_cluster_var_via_iv`. This function calculates the cluster variance by assuming an internal Inverse-Variance Parity (IVP) allocation among the assets in that branch, which avoids full matrix inversion and remains functional even with singular matrices.
3. **Weight Scaling:** Capital is split using the factor $\alpha = \sigma_R^2 / (\sigma_L^2 + \sigma_R^2)$ for the left branch and $1 - \alpha$ for the right.
4. **Termination:** The function recurses through the nested levels, scaling weights at each split, until `is_nested` returns false, indicating that individual assets have been reached.

3.3 Cluster Stability Across Rebalances

To assess how stable the partition produced by hierarchical clustering is over time, we compute the Rand index between the partitions obtained at consecutive rebalance dates, for cluster counts $k \in \{5, 8, 10, 15\}$, every covariance estimator and both detoning regimes. Figure 4 reports the 12-month EWM smoothed time series, while Table 5 summarises the long-run averages.

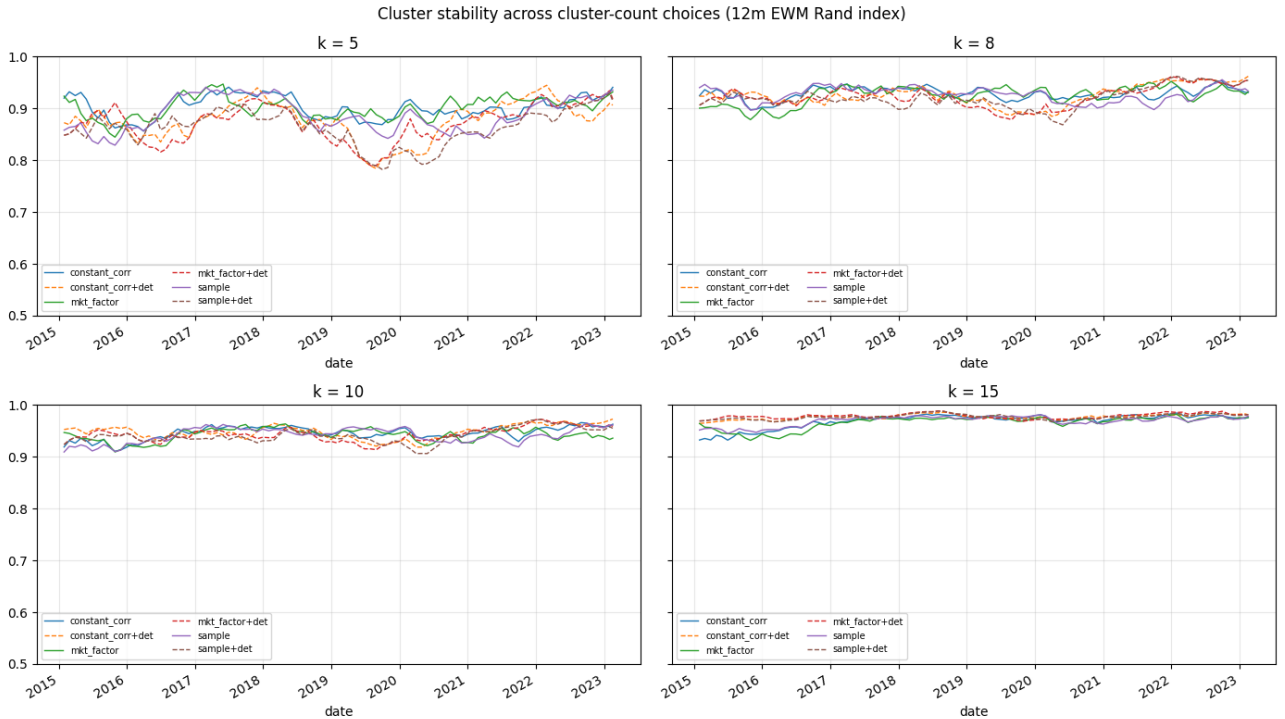


Figure 4: 12m EWM-smoothed Rand index between consecutive rebalances, by covariance estimator and detoning, for $k \in \{5, 8, 10, 15\}$. Solid lines: detoning off. Dashed lines: detoning on.

Estimator	Detone	$k = 5$	$k = 8$	$k = 10$	$k = 15$
constant_corr	False	0.904	0.929	0.948	0.972
constant_corr	True	0.878	0.928	0.949	0.978
mkt_factor	False	0.902	0.927	0.941	0.967
mkt_factor	True	0.876	0.925	0.945	0.979
sample	False	0.893	0.927	0.944	0.971
sample	True	0.866	0.923	0.943	0.978

Table 5: Average Rand index across all rebalances.

The results expose a clean separation between the two effects the assignment asks about, and they are not symmetric.

Effect of shrinkage. At every fixed value of k and detoning regime, the Rand index varies across the three estimators by less than ~ 0.04 , and at most cluster counts by less than 0.01. Moreover, the ordering among `sample`, `constant_corr` and `mkt_factor` is not even consistent across k : at $k = 5$ row `sample` is the worst, at $k = 8$ it is essentially tied with `constant_corr`. Hierarchical clustering depends on the ordering of pairwise distances, not on their magnitudes; shrinkage perturbs the magnitudes smoothly but largely preserves the ordering, so the dendrogram and its cuts are barely affected. The conclusion is that *shrinkage prior to clustering provides no meaningful additional benefit for cluster stability* – a sharp contrast with optimisation-based portfolios, where shrinkage is the main lever to control estimation noise.

Effect of detoning. The picture is more nuanced and depends on the granularity at which the dendrogram is cut:

- At low k ($k = 5$), detoning reduces stability by about 2–3 percentage points across all estimators. With only five clusters, the partition is dominated by the strongest separation in the data – which is precisely the market factor that detoning removes. The market signal is by far the most persistent component of the correlation structure, so the partition built on it is itself stable. Replacing it with weaker, noisier sectoral signals produces less stable clusters.
- At intermediate k ($k = 8, 10$) the two effects roughly cancel and the curves with and without detoning are essentially overlapping.
- At high k ($k = 15$) detoning improves stability by about 0.6–1.2 percentage points. At this granularity the partition is fine enough to resolve sectoral structure, and detoning makes those boundaries more evident.

3.4 Sensitivity of HRP Weights

We now study how HRP weights respond to the four design choices: linkage method, allocation traversal, covariance estimator, and detoning. The HRP procedure is run on the full grid `linkage` $\times$ `allocator` $\times$ `estimator` $\times$ `detoning` (48 configurations), and each variant is summarised by its mean rebalance-to-rebalance turnover $\langle \sum_i |\Delta w_i| \rangle$ and its mean Herfindahl concentration index $\langle \sum_i w_i^2 \rangle$. Figure 5 reports the two metrics averaged over estimators and detoning, isolating the linkage-allocator interaction.

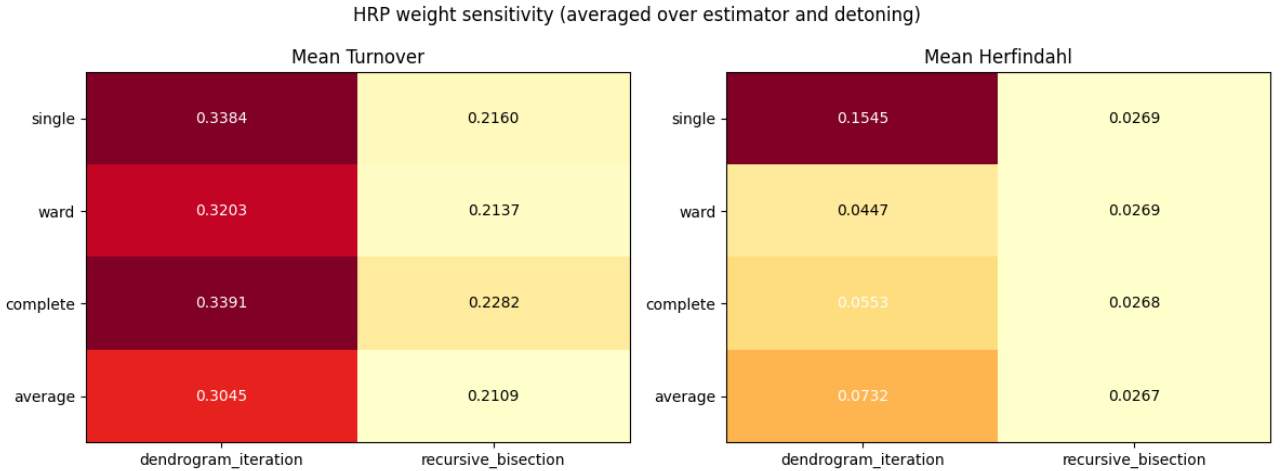


Figure 5: Mean turnover (left) and mean Herfindahl concentration (right) for each combination of linkage method and allocation traversal, averaged over the three covariance estimators and the two detoning regimes.

Two findings stand out. First, the choice of *allocator* dominates turnover: the dendrogram-iteration column is consistently higher than the recursive-bisection column, and this gap persists across every linkage method. Second, the linkage-allocator interaction is markedly asymmetric: under `recursive_bisection` the four linkage methods produce nearly identical Herfindahl indices, whereas under `dendrogram_iteration` the Herfindahl spans more than a factor of three, from 0.045 (`ward`) to 0.155 (`single`).

(i) How linkage propagates to the weights. The mechanism is the structural balance of the dendrogram. `single`-linkage trees suffer from the well-known chaining effect: pairs of close points are repeatedly absorbed into a single growing cluster, producing highly imbalanced trees with one deep, narrow branch and one short, wide branch. A top-down allocator that respects this hierarchy will allocate large amounts of capital to the deep branch and very little to the wide one, producing concentrated, high-turnover weights. `ward`-linkage trees, by

contrast, minimise within-cluster variance and produce balanced, symmetric trees that distribute capital more evenly; **complete** and **average** sit between these two extremes.

This is visible in Figure 6: under **dendrogram_iteration** the **single**-linkage portfolio (light blue) breaks away from the other three linkages by a wide margin, ending the period nearly 50% above the others. Under **recursive_bisection** the four lines collapse onto each other: the linkage choice becomes irrelevant.

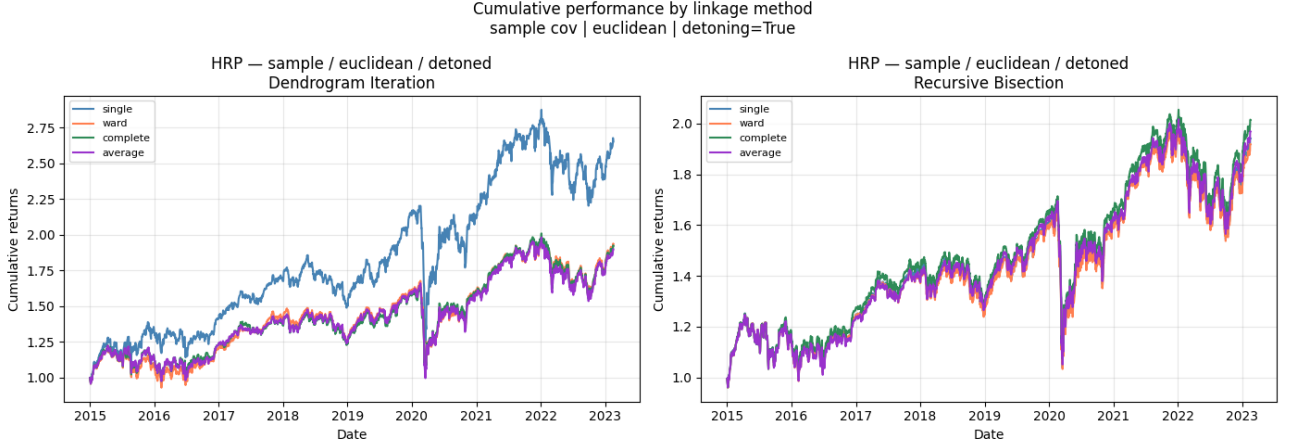


Figure 6: Cumulative performance by linkage method, with sample covariance, Euclidean second-stage distance and detoning enabled. Left: dendrogram iteration. Right: recursive bisection.

(ii) **Detoning – a small effect on weights.** For completeness, Figure 7 compares HRP cumulative performance with and without detoning, holding all other choices fixed. The two paths track each other closely, with detoning marginally outperforming under dendrogram iteration and marginally underperforming under recursive bisection. Although detoning produces an interpretably cleaner dendrogram, its effect on realised performance is second-order, consistent with the cluster-stability findings above.

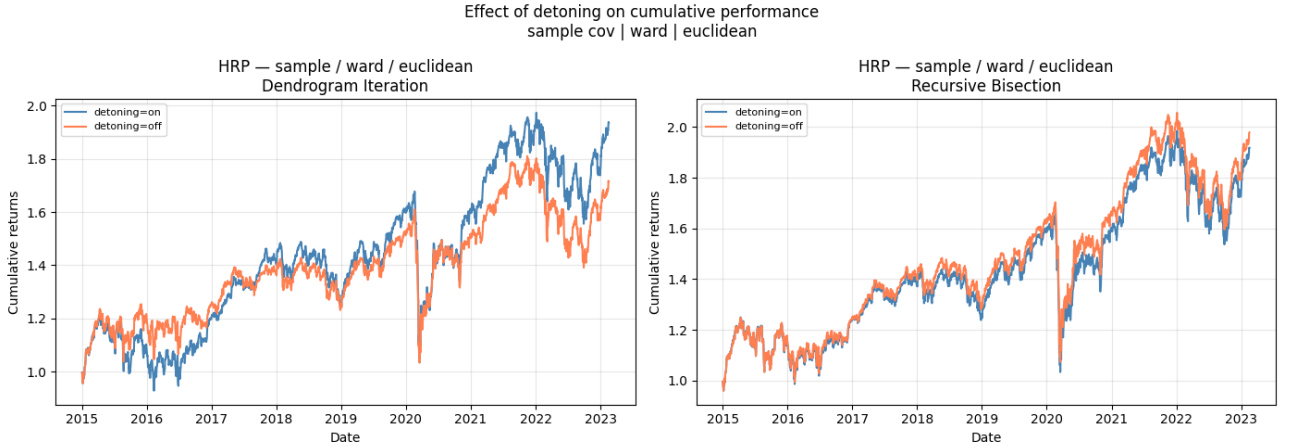


Figure 7: Effect of detoning on HRP cumulative performance, with sample covariance, ward linkage and Euclidean second-stage distance. Left: dendrogram iteration. Right: recursive bisection.

Summary. Among the four design choices, the linkage method is the dominant driver of HRP weights when paired with the dendrogram-iteration allocator; the choice of allocator itself controls how strongly any imbalance in the linkage structure propagates into concentrated weights; the estimator and detoning effects are an order of magnitude smaller. The combination **single** + **dendrogram_iteration** is the most stable and well-diversified configuration in our grid.

4 Part III – ERC vs HRP: Robustness and Performance Comparison

This final section compares the Equal Risk Contribution (ERC) portfolio with a representative Hierarchical Risk Parity (HRP) specification. The goal is dual: first, we assess how sensitive the two approaches are to

the covariance estimator; second, we compare their realised out-of-sample behaviour in terms of cumulative performance, realized volatility, drawdown and turnover.

The representative HRP strategy used in this section is based on the sample covariance matrix, Ward linkage, Euclidean distance, dendrogram iteration and correlation detoning.

4.1 Robustness to the covariance estimator

To evaluate robustness to the covariance estimator, we compute the average pairwise L^1 distance between portfolio weights obtained with different covariance estimators. For two portfolio weight vectors $w_t^{(a)}$ and $w_t^{(b)}$ at rebalance date t , the L^1 distance is:

$$d_t^{L^1}(a, b) = \sum_{i=1}^{N_t} |w_{i,t}^{(a)} - w_{i,t}^{(b)}|,$$

where the sum is computed on the common investable universe at that date. We then average this quantity over all rebalance dates.

Estimator pair	ERC mean L^1	HRP mean L^1
Sample vs constant-correlation	0.0000	0.2589
Sample vs market-factor	0.0000	0.1663
Constant-correlation vs market-factor	0.0000	0.2391
Average	0.0000	0.2214

Table 6: Average L^1 distance between weights across covariance estimators

The results show that ERC is essentially invariant to the choice of covariance estimator as shown above, while HRP is materially more sensitive.

HRP, conversly to ERC, depends on the full hierarchical structure extracted from the correlation matrix. Small changes in the estimated correlations may alter the dendrogram, the leaf ordering and the top-down allocation path. Therefore, even when the covariance estimates are close, the resulting HRP weights may differ more visibly.

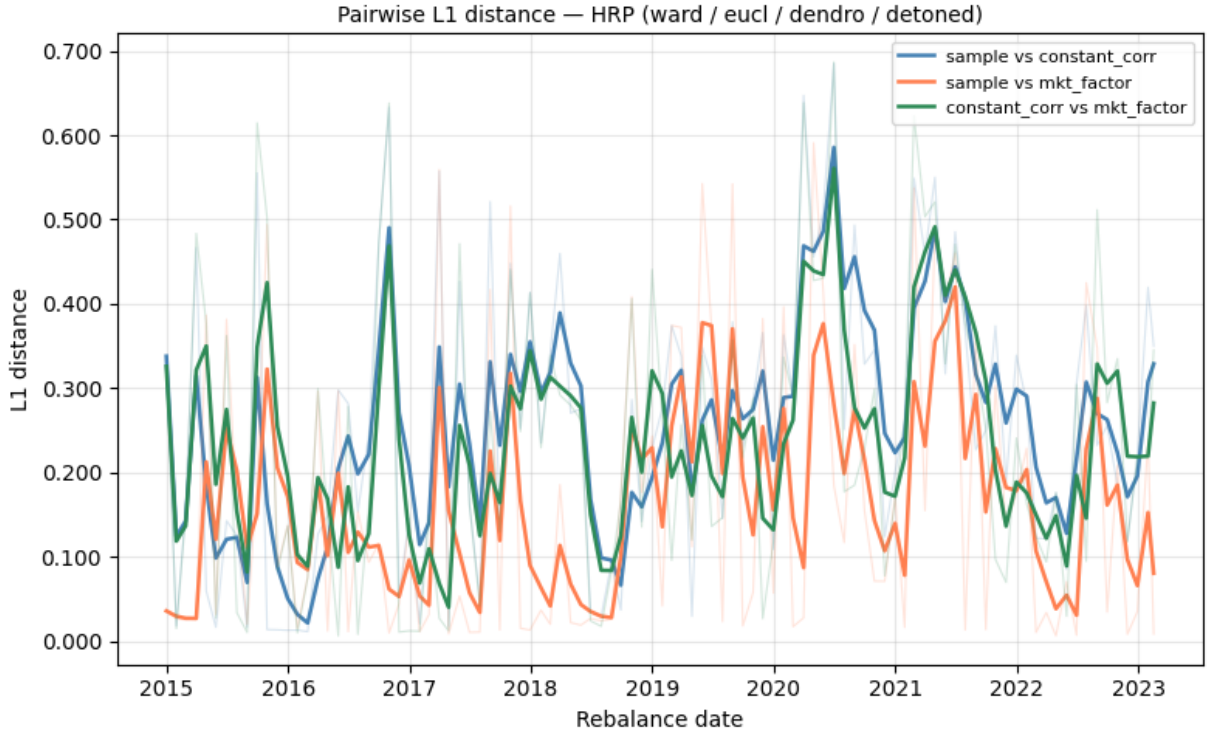


Figure 8: Pairwise L^1 distances between portfolio weights across covariance estimators

4.2 Head-to-head performance comparison

We then compare ERC based on the sample covariance matrix with the representative HRP configuration. Both strategies are rebalanced monthly and use the same rolling two-year estimation window. For each strategy we compute annualized return, annualized volatility, Sharpe ratio, maximum drawdown and average monthly turnover. The results are shown in the following table:

Strategy	Ann. Ret.	Ann. Vol.	Sharpe	Max DD	Avg. TO
ERC	8.75%	19.35%	0.452	-39.50%	0.0240
HRP	8.30%	18.93%	0.439	-38.29%	0.3567

Table 7: Performance comparison between ERC and representative HRP

The cumulative performance profiles are very close: ERC ends with a slightly higher annualised return, while HRP displays a slightly lower annualised volatility and a marginally less severe maximum drawdown. The Sharpe ratios are therefore very similar. From a pure risk-adjusted performance perspective, neither method strongly dominates the other in this sample.

The most relevant difference is the turnover. ERC has an average monthly turnover of only 0.0240, whereas HRP has an average monthly turnover of 0.3567. This means that the representative HRP allocation is substantially more active. The reason is structural: HRP weights depend on the estimated dendrogram, and changes in the clustering structure can generate discontinuous changes in allocation. ERC, by contrast, reacts more smoothly to changes in the covariance matrix because the weights are obtained from a continuous optimization problem.

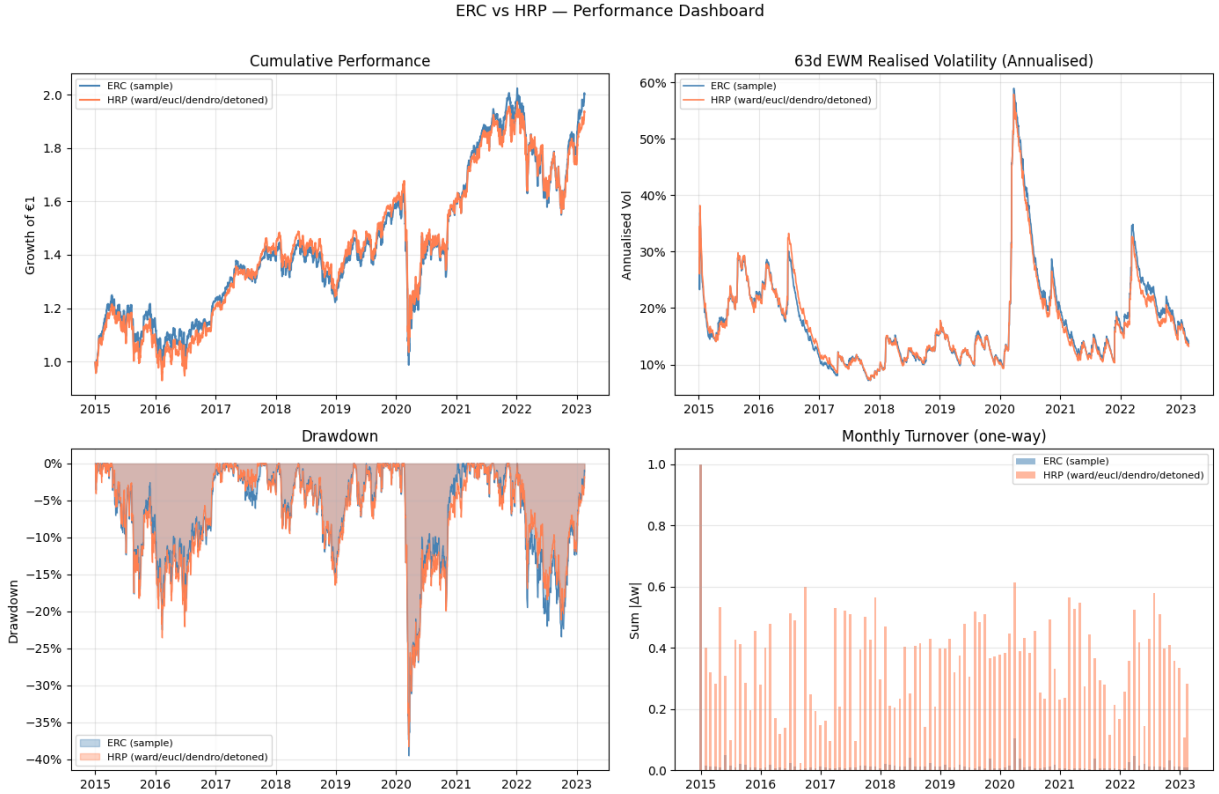


Figure 9: ERC versus HRP performance dashboard

4.3 Connecting with Lab 3: why shrinkage matters less

In Lab 3, shrinkage was introduced as a covariance estimation technique designed to reduce the noise of the sample covariance matrix. The main problem was that the sample covariance matrix estimates

$$\frac{N(N+1)}{2}$$

parameters, and therefore becomes noisy, ill-conditioned or even singular when the cross-sectional dimension is large relative to the available time series. The shrinkage estimator addresses this issue by combining the sample covariance matrix S with a structured target F :

$$\hat{\Sigma}_{\text{shrink}} = \delta F + (1 - \delta)S,$$

where the intensity δ balances estimation variance against target misspecification. The sample covariance is unbiased but high-variance, while the target is biased but stable; the optimal shrinkage intensity is therefore a bias–variance trade-off. In particular, shrinkage becomes more valuable when estimation error is large, for example when N is close to or larger than T , or when the effective sample size is small.

In the previous lab, this mattered directly because the portfolios considered there were minimum-variance and mean-variance portfolios. These methods use the covariance matrix in a very aggressive way, especially through matrix inversion. Small errors in $\hat{\Sigma}$ can therefore be amplified into unstable weights, large gross exposures and high turnover. So, in an environment like that, we expect shrinkage to have a visible impact: it improves conditioning, stabilizes the inverse covariance matrix and reduces the noise transmitted to the portfolio weights.

This assignment is however different. Both ERC and HRP are risk-based methods, they do not use expected returns, and they are less exposed to the classical instability of mean-variance optimization. ERC equalizes risk contributions subject to long-only and full-investment constraints, while HRP builds a hierarchical allocation from the correlation structure and avoids direct inversion of the covariance matrix. Therefore, the same covariance shrinkage techniques are still relevant, but their effect on realised performance is less pronounced here.

5 Comparison with Minimum Variance Portfolio

In this final stage of our analysis, we evaluate the practical advantages of the Hierarchical Risk Parity (HRP) approach by comparing it against the classical Minimum Variance (MV) portfolio. For this comparison, we selected a representative HRP configuration based on the following parameters: **sample covariance**, **Ward linkage**, **dendrogram iteration** as the allocation logic, and the application of **correlation detoning**.

The Minimum-Variance portfolio is obtained by solving the variance minimization objective, which yields the analytical solution:

$$w_{MV} = \frac{\Sigma^{-1}\mathbf{1}}{\mathbf{1}^T \Sigma^{-1} \mathbf{1}}$$

As discussed in the previous sections, this formulation relies explicitly on the inversion of the covariance matrix Σ , which could cause numerical errors and instability of results. Comparing with HRP, we do not expect dramatic differences in terms of raw performance metrics (such as cumulative returns or maximum drawdowns), instead, the primary objective of this comparison is to test the hypothesis of **superior weight stability** and **structural robustness**.

5.1 Comments on Results

The cumulative returns and drawdown profiles of the strategies are remarkably similar, with differences that are not statistically significant over the analyzed period. However, a deeper look at the operational metrics reveals some structural divergences.

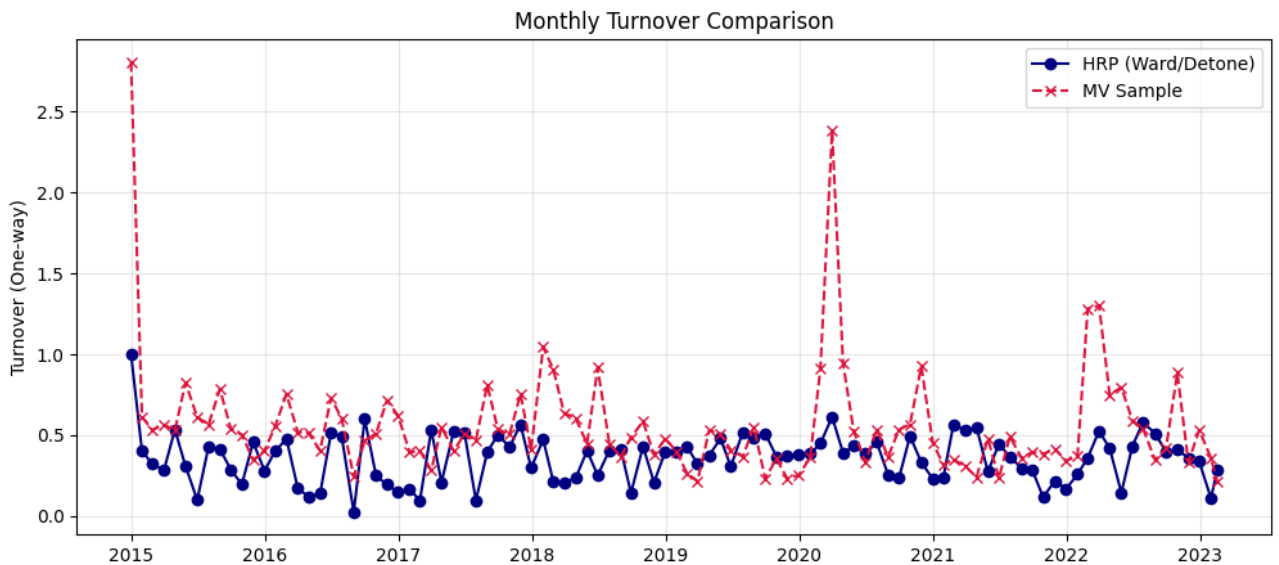


Figure 10: Turnover comparison MV vs HRP

The *monthly turnover* chart consistently shows that the Minimum Variance (MV) portfolio requires more frequent and substantial rebalancing compared to the HRP approach. When paired with the results from the Herfindahl Index, these findings strongly support our initial hypothesis: HRP provides superior weight stability and deeper diversification. While the MV portfolio tends to concentrate capital into a few low-volatility assets to minimize variance, HRP distributes risk more equitably across the entire cluster hierarchy.

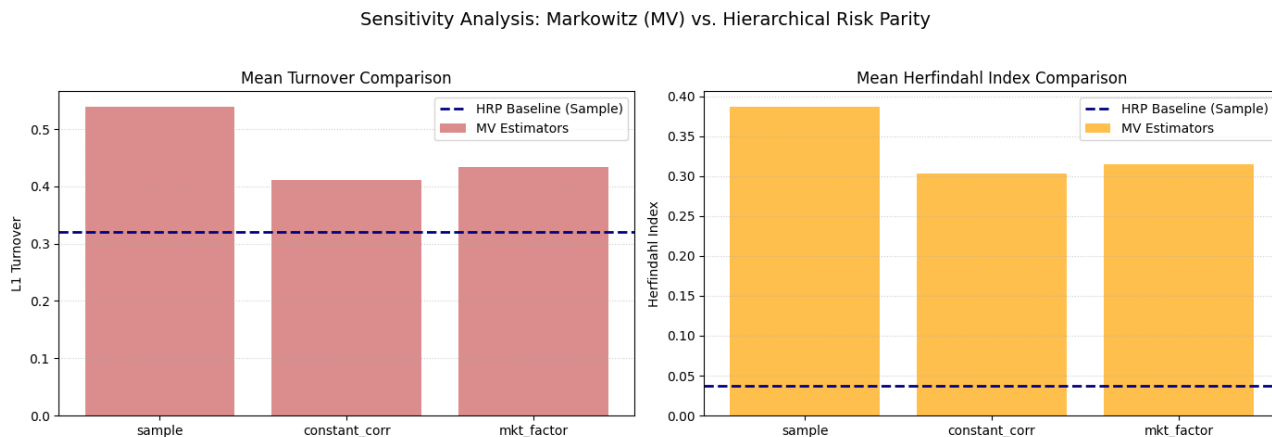


Figure 11: Mean Herfindahl index comparison

Furthermore, the comparative histogram confirms the effectiveness of *shrinkage* methods, as deeply discussed in previous assignments. Although these estimators significantly reduce turnover relative to the *sample* covariance matrix, the HRP approach maintains superior operational efficiency, proving to be intrinsically more robust against estimation noise.

6 Code Errors

The starter code provided with the assignment contained three deliberate bugs that we identified and corrected. Each is documented below.

Estimation window length

In the notebook, the rolling window for covariance estimation was set to `estimation_window = 252`, corresponding to a single year of trading days. The assignment specifies a *two-year* estimation window, so we corrected it to `estimation_window = 2 * 252`.

HRP correlation-to-distance map

The function `correlation_to_hrp_distance` in `hierarchical_risk_parity.py` returned

$$d_{ij} = \sqrt{2(1 + \rho_{ij})}$$

instead of the metric prescribed by López de Prado,

$$d_{ij} = \sqrt{0.5(1 - \rho_{ij})}.$$

Top-down allocation split factors

In the function `top_down_allocation`, the inverse-variance split between two sibling clusters was implemented as

```
alpha1 = cluster2_var * (cluster1_var + cluster2_var)
alpha2 = 1 + alpha1
```

The corrected version reads

```
alpha1 = cluster2_var / (cluster1_var + cluster2_var)
alpha2 = 1 - alpha1
```